Safer, Not Slower

Actionable insights from a large-scale survival analysis of baseball pitching mechanics

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Conference on Complex Systems 2026
Complexity & Sports Satellite



RICE UNIVERSITY
Sport Analytics

Collaborators

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Reboot Motion

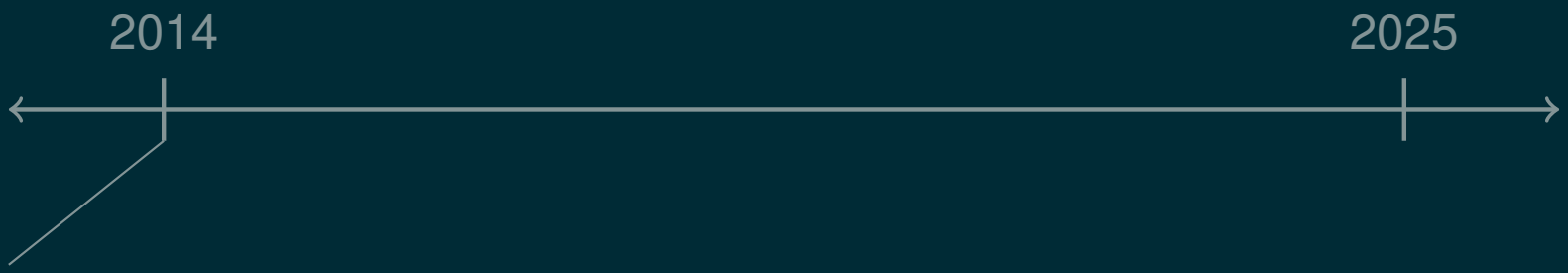
Daniel Barrueco
MLB

John D'Angelo
MLB

Amanda Glazer
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. . . and many others!

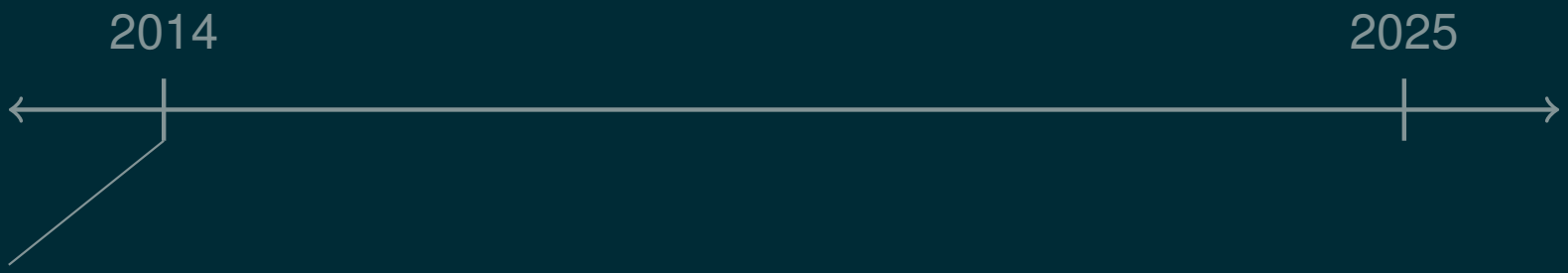
The Problem



"From my standpoint, [Tommy John surgery] is an **epidemic**."

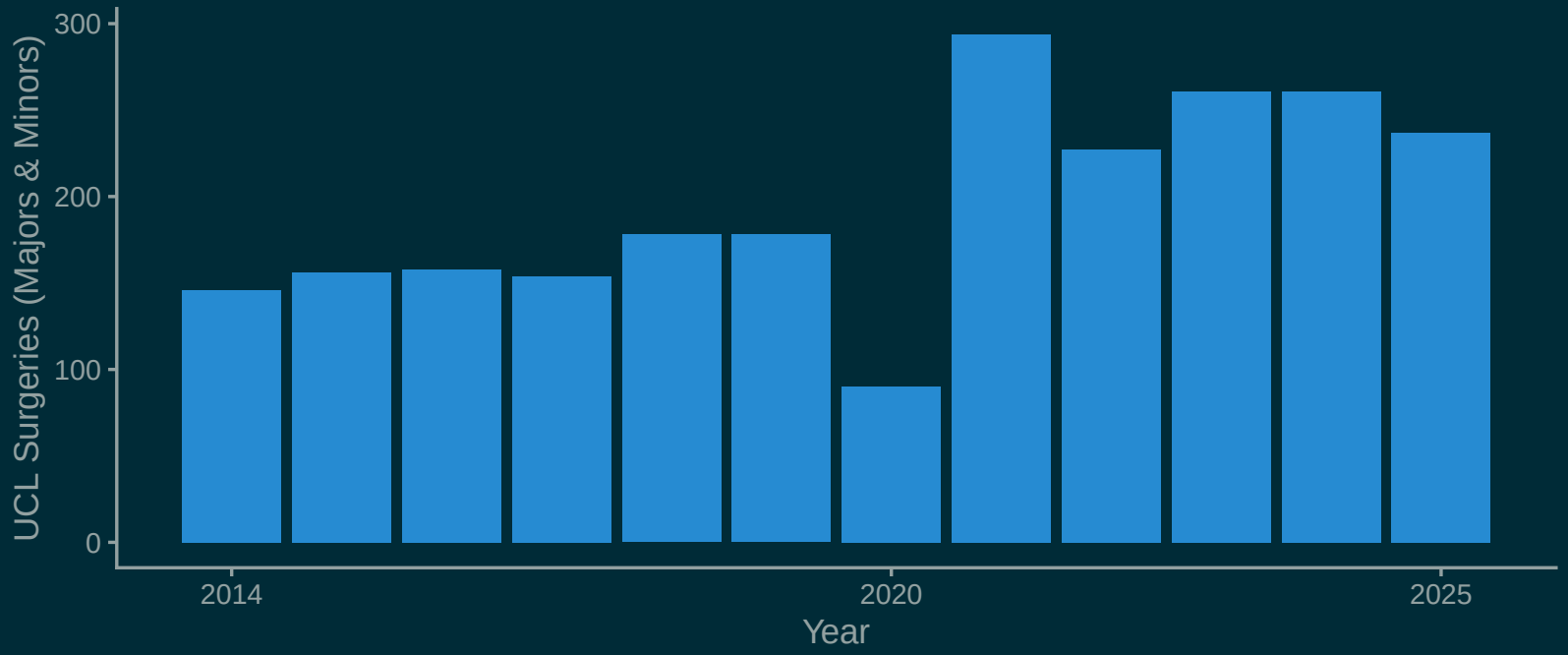
— Dr. James Andrews

The Problem

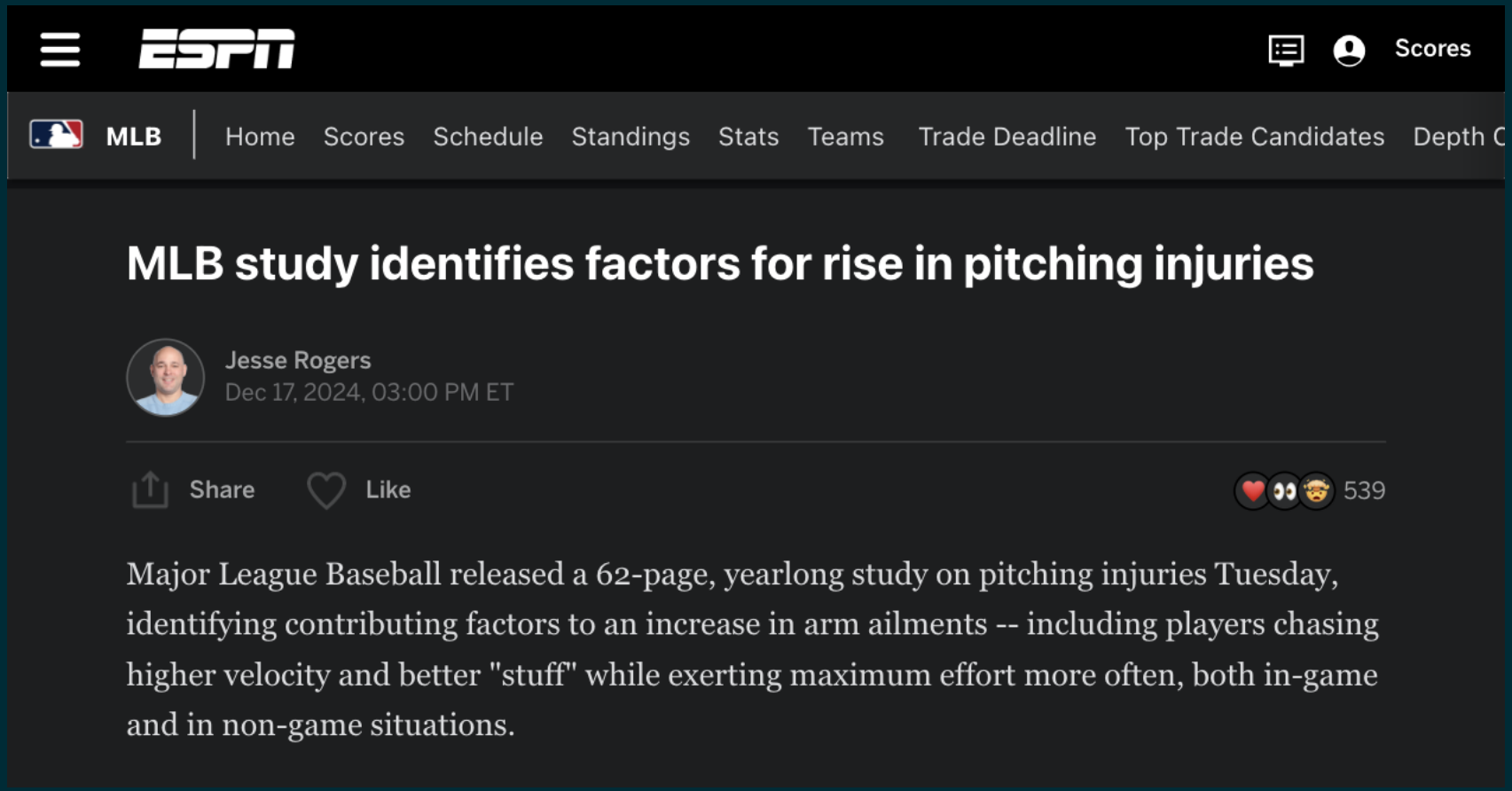


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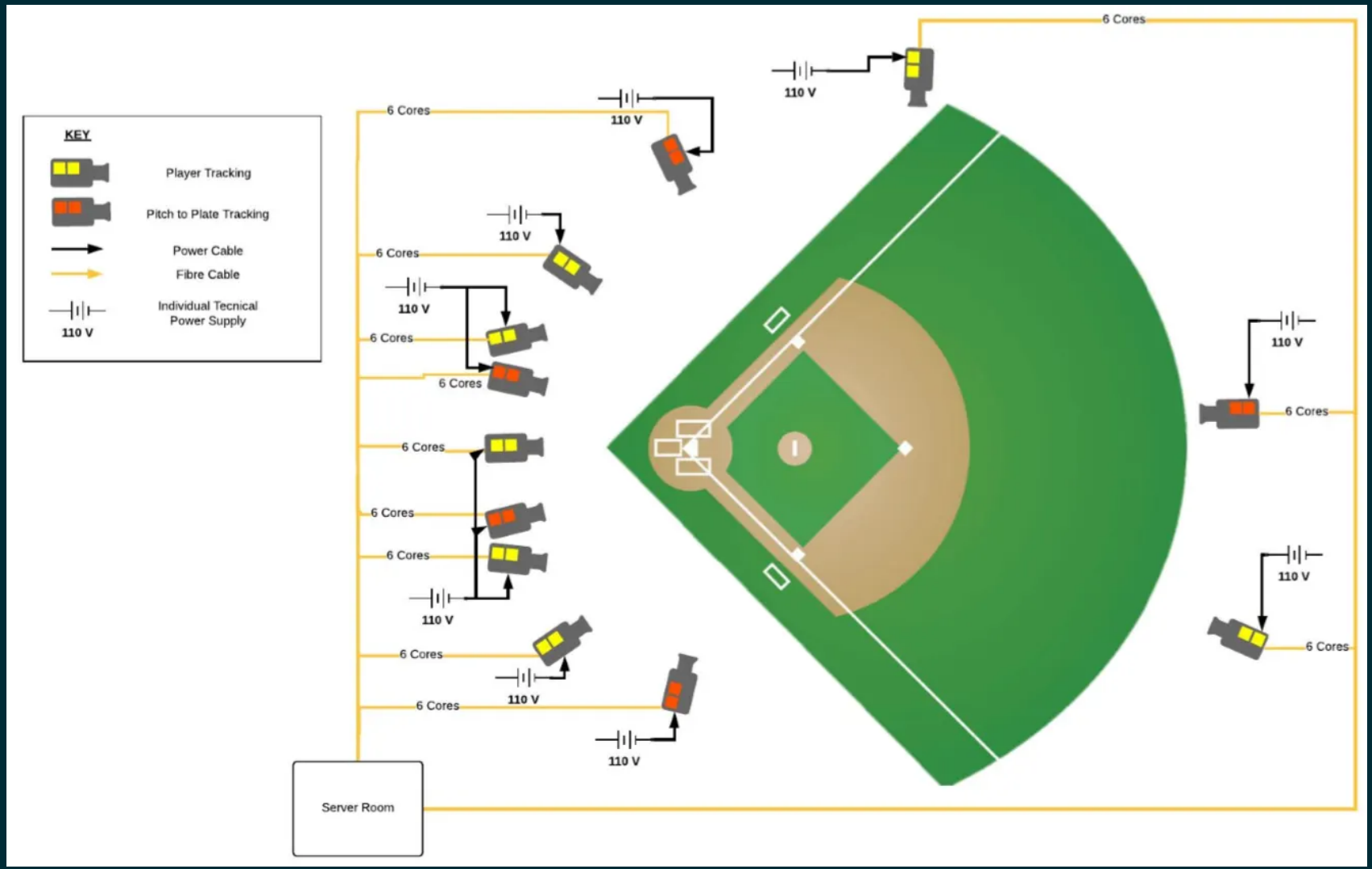
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2024 MLB Report



Data Generation



Data

Pitch-by-pitch:

- ball tracking, MLB 2017+ (5,972,263 pitches)
- ball tracking, minors 2021+ (11,322,383 pitches)
- **skeletal pose** tracking, MLB 2023+ (1,130,824 pitches)

Injury outcomes

- 8,234 throwing-arm IL placements, MLB & minors 2018+
- 1,726 (reported) UCL surgeries, MLB & minors 2018+

IL = injured list

UCL = ulnar collateral ligament

Methods

Framework:
year-Y features $\longmapsto$ year-$(Y+1)$ pitches until IL

Model: multivariate proportional hazards regression*

Cohorts:

- Majors 2018–2025 (3,828 pitcher-seasons)
- Minors 2022–2025 (6,241 pitcher-seasons)
- **Biomechanics 2024–2025** (1,211 pitcher-seasons)

*clustered standard errors account for repeated pitcher observations

Pitch-Tracking Features

Demographics

- Hand (L/R)
- Role (SP/RP)
- Past Injury (0/1)
- Age

Pitch Types

- SI Usage
- CT Usage
- SL+SW Usage
- CB+KC Usage
- CH+SF Usage

Pitch Tracking

- Primary Velo
- Primary Spin Rate
- Lateral Release Location
- FB Vert Break*
- SL Spin Rate*
- CB Spin Rate*
- Glove-Side Break
- Arm-Side Break
- Stuff+
- **Primary Velo Effort Ratio**

*interacted with usage

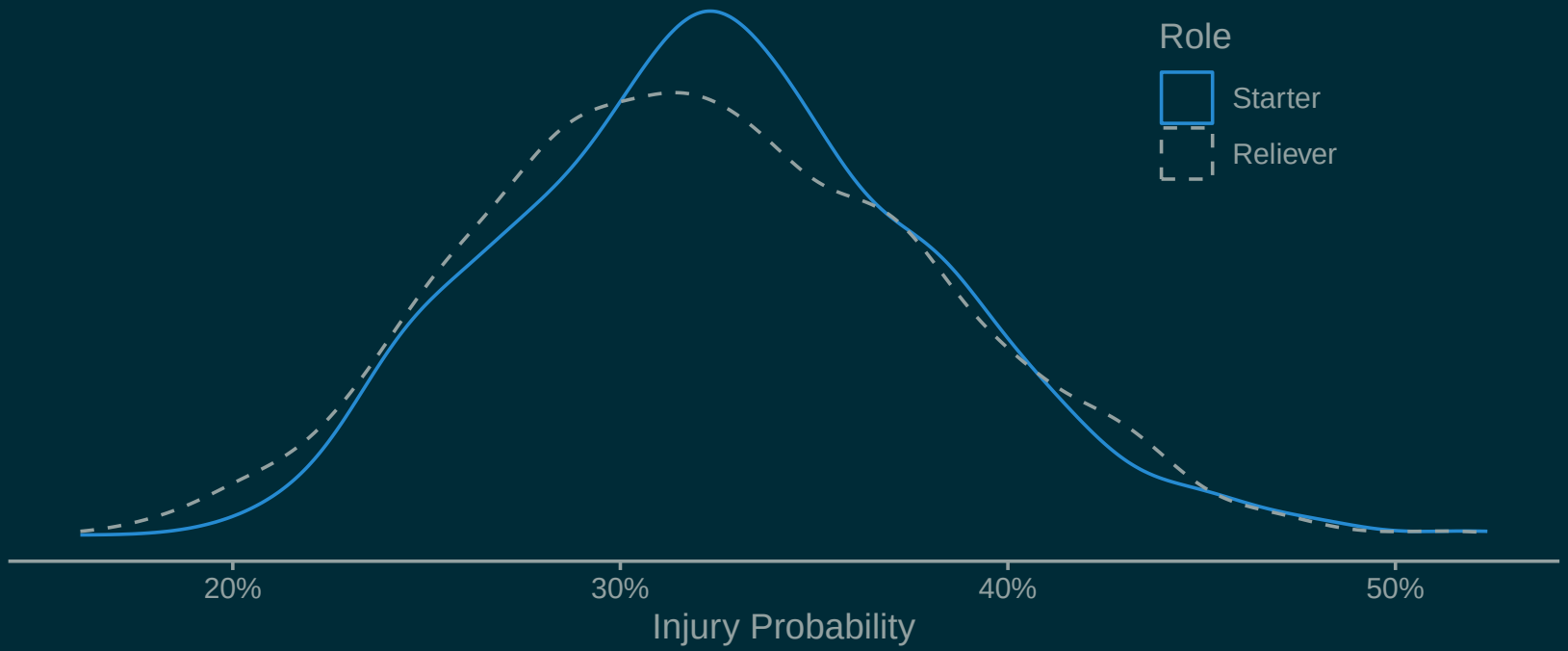
Pitch-Tracking Results

	95% Conf. Int.
Majors 2018–2025 cohort	
• 1 mph Primary Velo $\leftrightarrow +5\%$ hazard	$(+2\%, +8\%)$
• **10pp SL+SW Usage $\leftrightarrow +8\%$ hazard**	$(+1\%, +15\%)$
• 1 SD FB Vert Break $\leftrightarrow +10\%$ hazard	$(+3\%, +18\%)$
Minors 2022–2025 cohort	
• 1 mph Primary Velo $\leftrightarrow +5\%$ hazard	$(+2\%, +8\%)$
• **10pp SL+SW Usage $\leftrightarrow +6\%$ hazard**	$(+0\%, +13\%)$
• 10pp CH+SF Usage $\leftrightarrow +8\%$ hazard	$(+1\%, +16\%)$
• 1 SD Lat. Rel. Position $\leftrightarrow -7\%$ hazard	$(-12\%, -2\%)$

SL+SW = slider/sweeper
CH+SF = changeup/splitter

pp = percentage point
SD = standard deviation

Compounding Risk



Fitted injury probability* **varies greatly** across pitcher-seasons

*Reflects 2,500-pitch starter workload or 1,000-pitch reliever workload

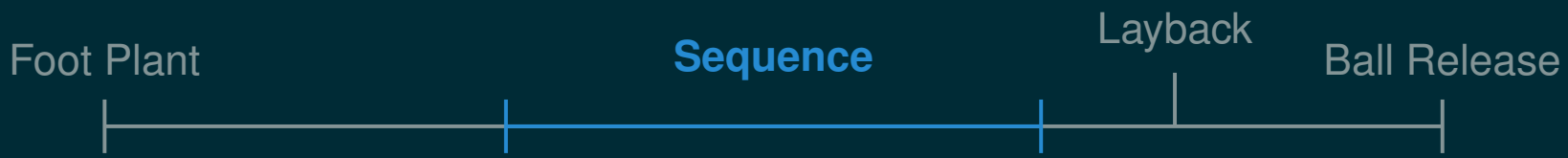
Biomechanics Features (Joint Angles)

Joints:

- Rear Hip (3 df)
- Lead Knee (1 df)
- Lumbar (3 df)
- Shoulder (3 df)
- Elbow (1 df)

Summary stats:

- min/max
- min/max timing
- min/max in **sequence**
- range utilization
- at layback

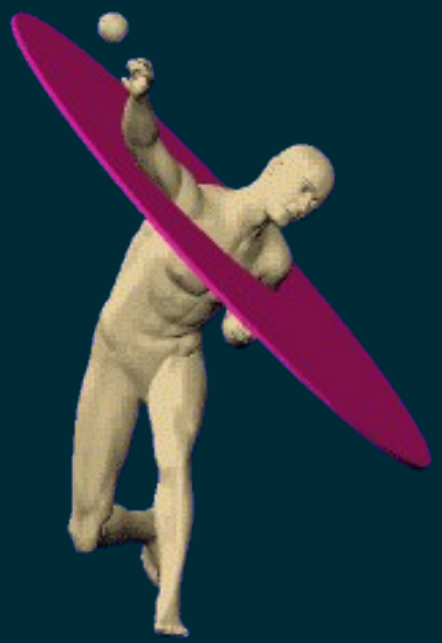


$$\text{range utilization} = \frac{\text{range in sequence}}{\text{full range}}$$

df = degrees of freedom

67 features

Biomechanics Features (Momentum)



from Nyman (2008)

Body segments:

- Rear Leg
- Lead Leg
- Glove Arm
- Torso
- Upper Arm
- Forearm
- Hand

Summary stats:

- tilt of rotation plane
- angular momentum projected onto plane of pitching hand (min/max, at release)
- angular momentum orthogonal to pitching hand plane (min/max)

67 features + **72** features = 139 features

Biomechanics Suspects

1. Max Elbow Varus Torque
2. Arm Slot
3. Min Shoulder Horizontal Abduction* (aka scap loading)
4. Max Glove Arm Momentum* (projected onto pitching hand)
5. Max Torso Momentum* (projected onto pitching hand)

Model: multivariate proportional hazards regression

- include demographic and significant pitch-tracking features

*previously implicated at Saberseminar! (separate dataset)

Biomechanics Results

Feature	Effect*	95% Conf. Int.
Max Elbow Varus Torque	+2%	(–8%, +14%)
Arm Slot	+7%	(–7%, +22%)
Min Shoulder Horizontal Abd.	–4%	(–14%, +6%)
Max Glove Arm Momentum	**–21%**	**(–29%, –12%)**
Max Torso Momentum	+3%	(–8%, +16%)

*corresponding to 1-standard-deviation increase in feature

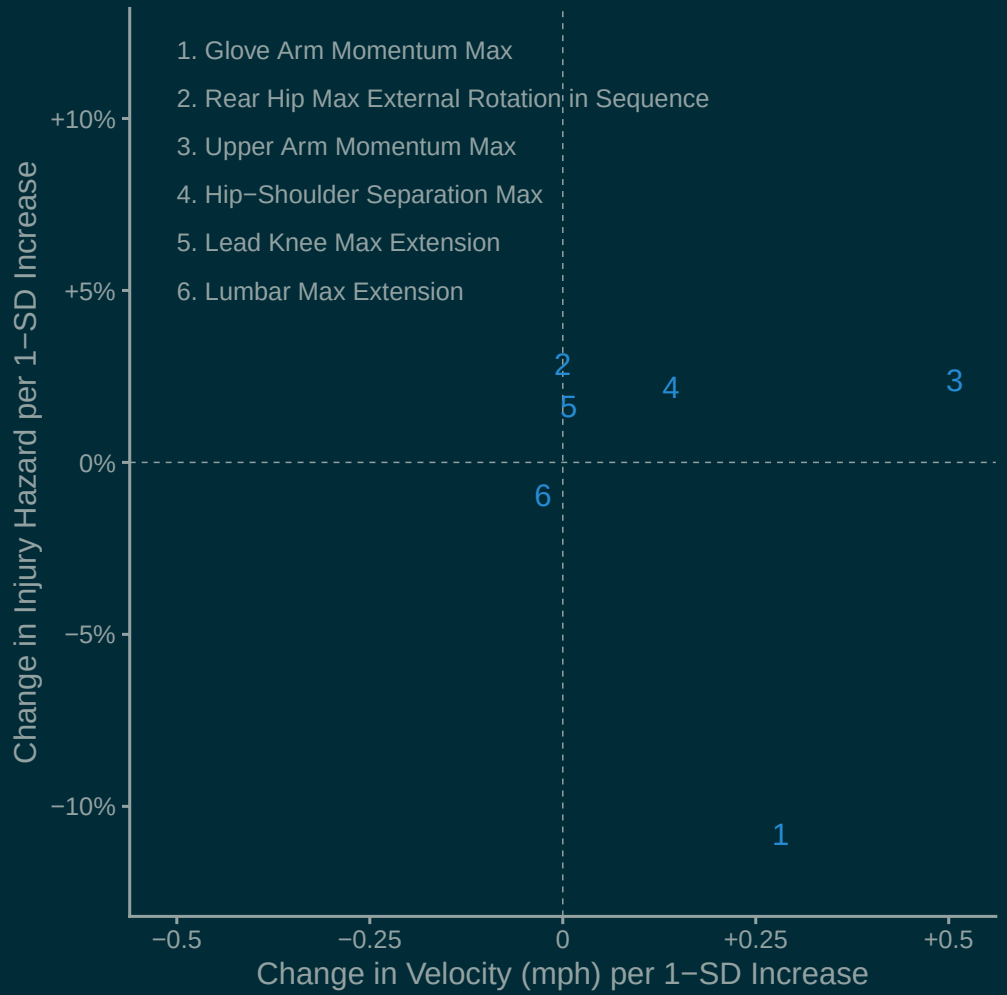
Data Mining

Model: lasso regression

- include demographic and significant pitch-tracking features
- include all 139 biomechanics features
- ℓ_1 **penalty** achieves variable selection

Two outcomes: (1) pitches until IL and (2) fastball velocity

Data Mining Results



Glove Arm Momentum Max is the only metric associated with **increased** velocity and **decreased** injury hazard

What have we learned?

First public study to estimate pitcher injury survival model using pitch-tracking data (**20-50%** injury probability over full season)

Non-fastball usage (especially **slider/sweeper**) is positively associated with injury hazard in both MLB and minors

Glove arm momentum is strongly predictive of reduced injury hazard (and increased velocity), warranting further research

Notable negative findings

- Primary Velo Effort Ratio
- Spin Rate and Glove-Side Break

Bayesian Changepoint Detection

Project led by **Rose Graves** (Rice Stats PhD student)

Goal: Identify when pitchers are pitching fatigued

Problem:

- Measurements are noisy
- Fatigue onset by game likely similar but not identical

Solution: Bayesian Hierarchical Changepoint model

1. Hierarchy across games within pitcher
2. Hierarchy across pitchers within league

Changepoint Causal Inference

Project led by **Amanda Glazer** (University of Texas)

Goal: Estimate causal effect of velocity increase

Problem:

- Rolling “enrollment” in treatment
- No treatment labels in the data

Solution:

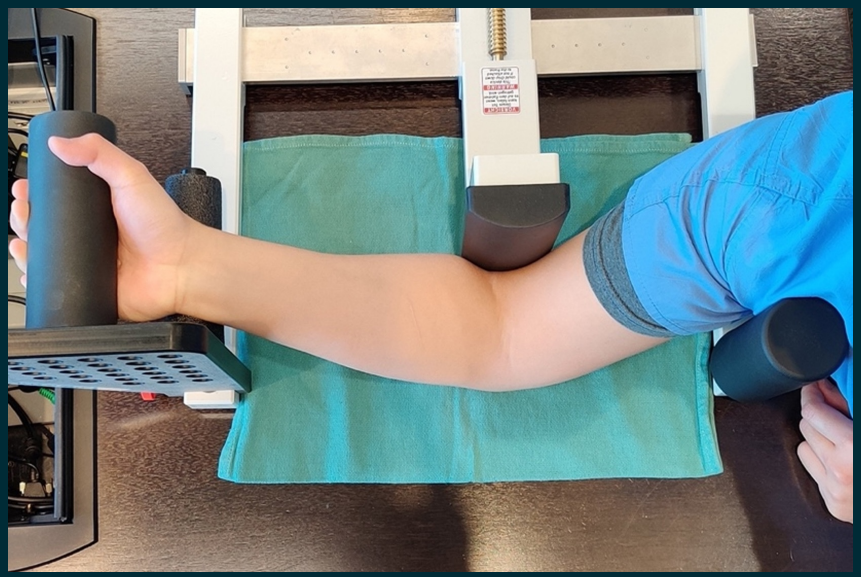
1. Identify statistically significant changepoints
2. Perform rolling treatment-control matching

Shear Wave Elastography

Collaboration with **Dr. Robert Jack** (Houston Methodist)

Problem: Not enough pitching injuries!

Solution: Measure laxity of UCL after pitching (pilot study)



from Gupta et al. (2023)

Thank You!

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